

大学化学期中复习配套考点题目（致新书院）

ZJJ（配套课件笔记使用）专一针对大学化学，化学原理的朋友也可快速扫描

宗旨是帮大家取得一个理想的分数（未完待续）

大学化学期中复习配套考点题目（致新书院）

概述

Ch 1

- Point 1（均一/不均一）
- Point 2（精确度/准确度）
- Point 3（物理/化学变化）
- Point 4（混合物分离的简单过程）
- Point 5（有效数字）

Ch 2

- Point 1（道尔顿四条假设/原子结构的发现）
- Point 2（常见族的英文名称）
- Point 3（卢瑟福粒子散射实验）
- Point 4（原子结构）
- Point 5（命名----非常重要）

Ch 3

- Point 1（反应名称及区分）
- Point 2（方程式配平）
- Point 3（特殊爱考方程）

Ch 4

- Point 1（旁观离子）
- Point 2（电解质概念）
- Point 3（M的各种题目-非常重要）
- Point 4（关于水溶液中方程式的不同问法）

概述

半期考试题型为Part A（选择题），Part B（简答题），Part C（大题）

Part A为20-25选择题（2分一道）

Part B为5-6道简答题（4-6分一道）

Part C为3-4道计算题（8-12分一道）

考试卷子后面附带的元素周期表很难随时翻页查阅，可以考场问老师能否从试卷撕下来方便随时查看（上学期是可以的）

（余老师班）

LV19 管理员 助教



但是咱们班有两名同学很可惜，没有看到最后一部分大题，导致26分没有写。下次考试时一定记得不要缺答漏答。

你如果特别想知道，那就这两天找个时间来我办公室看你的答题卡

但是不能带走，不能拍照哦

噢噢好的 明天早上可以来吗

明天上午10点到11点半之间吧！来创园5栋606

Outline

Chapter 1: Introduction: Matter and Measurement

Chapter 2: Atoms, Molecules, and Ions

Chapter 3: Chemical Reactions and Reaction Stoichiometry

Chapter 4: Reactions in Aqueous Solution

Chapter 5: Thermochemistry

Chapter 6: Electronic Structure of Atoms

Chapter 7: Periodic Properties of the Elements

Ch 1

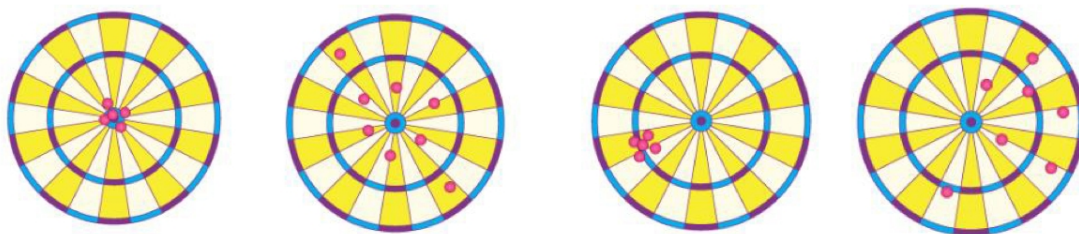
Point 1 (均一/不均一)

7. Which of the following is a homogeneous mixture?

A) blood B) copper wire C) steel D) fresh onion soup E) muddy water

合金都是均质

Point 2 (精确度/准确度)



Point 3 (物理/化学变化)

- (a) striking a match causes it to burst into flame.
- (b) Water vapor in the air forms frost.
- (c) rusting of an iron bridge.

A3) In the following list, only _____ is chemical change.

- A) a wet towel dries in the sun
- B) dissolving sugar in water
- C) the density of gold is 19.3 g/mL
- D) an antacid tablet decreases the acidity of an over acid stomach
- E) compression of oxygen gas

容易出现的是些固液气状态变化过程的干扰

Point 4 (混合物分离的简单过程)

A1) Which one of the following is often easily separated into its components by simple techniques such as filtering or decanting?

- A) heterogeneous mixture
- B) compounds
- C) homogeneous mixture
- D) elements
- E) solutions

Point 5 (有效数字)

Don't round-up your data until the final result

Mass from ^{12}C atoms

+ Mass from ^{13}C atoms

$$12 \times 0.9889$$

(4 Sig Fig)

+

$$13 \times 0.0111$$

(3 Sig Fig)

$$11.8668$$

+

$$0.1443 = 12.0111 \approx 12.01$$

↑
Estimated, 2 decimal place

↑
Estimated

↑
2 decimal place

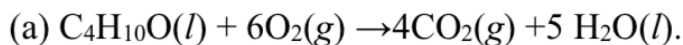
A4) Which of the following has the same number of significant figures as the number 1.00310?

- A) 1×10^6
- B) 5.119
- C) 199.791
- D) 100
- E) 8.66

下面这道题涉及的有效数字极其抽象

17. Diethyl ether, $\text{C}_4\text{H}_{10}\text{O}(l)$, a flammable compound that has long been used as a surgical anesthetic, has the structure $\text{H}_3\text{C}-\text{CH}_2-\text{O}-\text{CH}_2-\text{CH}_3$. The complete combustion of 1 mol of $\text{C}_4\text{H}_{10}\text{O}(l)$ to $\text{CO}_2(g)$ and $\text{H}_2\text{O}(l)$ yields $\Delta H^\circ = -2723.7 \text{ kJ}$. (a) Write a balanced equation for the combustion of 1 mol of $\text{C}_4\text{H}_{10}\text{O}(l)$. (b) By using the information in this problem and data below, calculate ΔH_f° for diethyl ether.

Substance	ΔH_f° (kJ/mol)
$\text{CO}_2(g)$	-393.5
$\text{O}_2(g)$	0
$\text{H}_2\text{O}(g)$	-241.82
$\text{H}_2\text{O}(l)$	-285.83



(b) $\Delta H_{\text{rxn}} = 4\Delta H_f^\circ[\text{CO}_2(g)] + 5\Delta H_f^\circ[\text{H}_2\text{O}(l)] - 6\Delta H_f^\circ[\text{O}_2(g)] - \Delta H_f^\circ[\text{C}_4\text{H}_{10}\text{O}(l)]$
 $-2723.7 = 4 \times (-393.5) + 5 \times (-285.83) - 0 - \Delta H_f^\circ[\text{C}_4\text{H}_{10}\text{O}(l)]$

Ch 2

Point 1 (道尔顿四条假设/原子结构的发现)

1. Which one of the following is not one of the postulates of Dalton's atomic theory?
- A) All atoms of a given element are identical; the atoms of different elements are different and have different properties.
 - B) Each element is composed of extremely small particles called atoms.
 - C) Atoms of an element are not changed into different types of atoms by chemical reactions: atoms are neither created nor destroyed in chemical reactions.
 - D) Atoms are composed of protons, neutrons, and electrons.
 - E) Compounds are formed when atoms of more than one element combine; a given compound always has the same relative number and kind of atoms.

大致记住中文观点，之后进行排除选择

Point 2 (常见族的英文名称)

(描述型)

15. _____ are found uncombined, as monatomic species in nature.
- A) Noble gases
 - B) Chalcogens
 - C) Alkali metals
 - D) Alkaline earth metals
 - E) Halogens

(直接型)

- A7) The elements in groups 1A, 6A, and 7A are called
- A) alkaline earth metals, halogens, and chalcogens
 - B) alkali metals, chalcogens, and halogens
 - C) alkali metals, halogens, and noble gases
 - D) alkaline earth metals, transition metals, and halogens
 - E) halogens, transition metals, and alkali metals

还有可能在题目中直接出现，所以这个一定要记住。

Point 3 (卢瑟福粒子散射实验)

4. Cathode rays are
- A) neutrons
 - B) X-rays
 - C) electrons
 - D) protons
 - E) atoms

Point 4 (原子结构)

A6) An elemental ion has 22 protons, 20 electrons, and 26 neutrons. Which isotope is it?

- A) ${}^{47}_{26}\text{Fe}$ B) ${}^{48}_{26}\text{Fe}$ C) ${}^{48}_{22}\text{Ti}$
D) ${}^{48}_{22}\text{Ti}^{2+}$ E) ${}^{48}_{22}\text{Ti}^{2-}$

Point 5 (命名----非常重要)

Which name is **incorrectly** paired with the formula?

- a. hydroxide ion; OH^-
- b. barium(II) chloride; BaCl_2
- c. ammonia; NH_3
- d. sulfur dioxide; SO_2

➤ Common Cations

TABLE 2.4 • Common Cations*

Charge	Formula	Name	Formula	Name
1+	H ⁺	hydrogen ion	NH ₄ ⁺	ammonium ion
	Li ⁺	lithium ion	Cu ⁺	copper(I) or cuprous ion
	Na ⁺	sodium ion		
	K ⁺	potassium ion		
	Cs ⁺	cesium ion		
	Ag ⁺	silver ion		
2+	Mg ²⁺	magnesium ion	Co ²⁺	cobalt(II) or cobaltous ion
	Ca ²⁺	calcium ion	Cu ²⁺	copper(II) or cupric ion
	Sr ²⁺	strontium ion	Fe ²⁺	iron(II) or ferrous ion
	Ba ²⁺	barium ion	Mn ²⁺	manganese(II) or manganous ion
	Zn ²⁺	zinc ion	Hg ₂ ²⁺	mercury(I) or mercurous ion
	Cd ²⁺	cadmium ion	Hg ²⁺	mercury(II) or mercuric ion
			Ni ²⁺	nickel(II) or nickelous ion
			Pb ²⁺	lead(II) or plumbous ion
3+	Al ³⁺	aluminum ion	Sn ²⁺	tin(II) or stannous ion
			Cr ³⁺	chromium(III) or chromic ion
			Fe ³⁺	iron(III) or ferric ion

➤ Common Anions

TABLE 2.5 • Common Anions*

Charge	Formula	Name	Formula	Name
1−	H [−]	hydride ion	CH ₃ COO [−]	acetate ion
			(or C ₂ H ₃ O ₂ [−])	
	fluorine F [−]	fluoride ion	ClO ₃ [−]	chlorate ion
	chlorine Cl [−]	chloride ion	ClO ₄ [−]	perchlorate ion
	bromine Br [−]	bromide ion	NO ₃ [−]	nitrate ion
	iodine I [−]	iodide ion	MnO ₄ [−]	permanganate ion
	CN [−]	cyanide ion		
2−	OH [−]	hydroxide ion		
	O ^{2−}	oxide ion	CO ₃ ^{2−}	carbonate ion
	O ₂ ^{2−}	peroxide ion	CrO ₄ ^{2−}	chromate ion
	Sulfur S ^{2−}	sulfide ion	Cr ₂ O ₇ ^{2−}	dichromate ion
3−			SO ₄ ^{2−}	sulfate ion
	N ^{3−}	nitride ion	PO ₄ ^{3−}	phosphate ion

*The ions we use most often are in boldface. Learn them first.
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建议考前拿着元素周期表整体过一遍（与自己所想的变形进行比较而不是仅知道变形规则）

自问环节：



（共价化合物or离子化合物）

我难大家难。

Ch 3

Point 1 (反应名称及区分)

复分解/中和反应 metathesis/neutralization reactions

是否能区分各个反应

例如：化合反应是否必须要求反应物只有两个？

分解反应是否必须要求生成物只有两个？

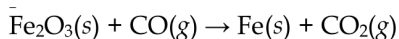
燃烧反应是否必须要求有氧气参与？ (or绝大多数?)

是否在化学反应方程式中写了物质的状态

Point 2 (方程式配平)

方程的配平：

升失氧，降得还。

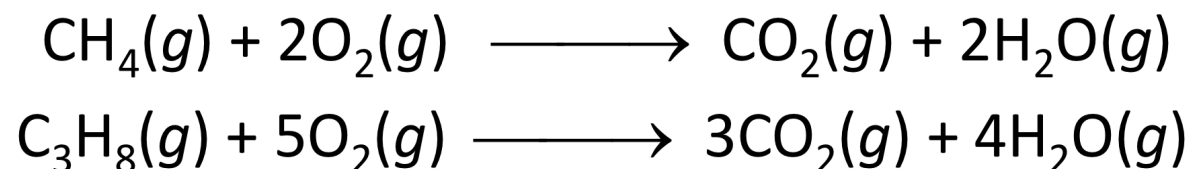


(a) Balance this equation.

关于燃烧反应 (C, H, O)

- 1.直接由反应物的C个数写出二氧化碳前的系数，直接由反应物O的个数写出
- 2.直接由反应物的C个数写出水前的系数，直接由反应物H的个数写出 (注意一半)
- 3.最后反推氧气的系数
- 4.有分数等最后翻倍

举例：



Point 3 (特殊爱考方程)



Ch 4

Point 1 (旁观离子)

3. What are the **spectator** ions in the reaction between $\text{KOH}(aq)$ and $\text{HNO}_3(aq)$?
- A) K^+ and H^+
B) H^+ and OH^-
C) K^+ and NO_3^-
D) H^+ and NO_3^-
E) OH^- only

Point 2 (电解质概念)

两种问法

1. A weak electrolyte exists predominantly as _____ in solution.
A) an isotope B) molecules C) ions D) atoms E) electrons
2. Which of the following would you least expect to find in $\text{CaCl}_2(aq)$?
A) H_2O molecules
B) CaCl_2 molecules
C) Ca^{2+} ions
D) Cl^- ions
E) Ca^{2+} and Cl^- ions

Point 3 (M的各种题目-非常重要)

11. The concentration of species in 500 mL of a 2.104 M solution of sodium sulfate is _____ M sodium ion and _____ M sulfate ion.
A) 1.052, 1.052
B) 4.208, 2.104
C) 2.104, 4.208
D) 2.104, 2.104
E) 2.104, 1.052
8. A 0.355 M K_2SO_4 solution can be prepared by _____.
A) dilution of 500.0 mL of 1.00 M K_2SO_4 to 1.00 L
B) dissolving 46.7 g of K_2SO_4 in water and diluting to a total volume of 250.0 mL
C) dissolving 46.7 g of K_2SO_4 in water and diluting to 500.0 mL, then diluting 25.0 mL of this solution to a total volume of 500.0 mL
D) dilution of 1.00 mL of 0.355 M K_2SO_4 to 1.00 L
E) diluting 46.7 mL of 1.90 M K_2SO_4 to 250.0 mL
13. All of the following are true concerning 2.00 L of 0.100 M solution of Na_3PO_4 **except** for _____.
A) This solution contains 0.200 mol of Na_3PO_4 .
B) 1.00 L of this solution contains 0.300 mol of Na^+ ions.
C) There are 3.01×10^{22} phosphorus atoms in 500.0 mL of this solution.
D) This solution contains 0.200 mol of oxygen atoms.
E) This solution contains 0.600 mol of Na^+ ions.
17. (6 points) A solution is made by mixing 15.0 g of $\text{Sr}(\text{OH})_2$ and 55.0 mL of 0.200 M HNO_3 . Neglect volume changes.
(a) Write a balanced equation for the reaction that occurs between the solutes.
(b) Calculate the concentration of each ion remaining in solution.
(c) Is the resultant solution acidic or basic? Why?

Point 4 (关于水溶液中方程式的不同问法)

12. The net ionic equation for formation of an aqueous solution of $\text{Al}(\text{NO}_3)_3$ via mixing solid $\text{Al}(\text{OH})_3$ and aqueous nitric acid is _____.

- A) $\text{Al}(\text{OH})_3(s) + 3 \text{NO}_3^-(aq) \rightarrow 3 \text{OH}^-(aq) + \text{Al}(\text{NO}_3)_3(s)$
- B) $\text{Al}(\text{OH})_3(s) + 3 \text{HNO}_3(aq) \rightarrow 3 \text{H}_2\text{O}(l) + \text{Al}^{3+}(aq) + 3 \text{NO}_3^-(aq)$
- C) $\text{Al}(\text{OH})_3(s) + 3 \text{H}^+(aq) \rightarrow 3 \text{H}_2\text{O}(l) + \text{Al}^{3+}(aq)$
- D) $\text{Al}(\text{OH})_3(s) + 3 \text{NO}_3^-(aq) \rightarrow 3 \text{OH}^-(aq) + \text{Al}(\text{NO}_3)_3(aq)$
- E) $\text{Al}(\text{OH})_3(s) + 3 \text{HNO}_3(aq) \rightarrow 3 \text{H}_2\text{O}(l) + \text{Al}(\text{NO}_3)_3(aq)$